

View Review

Paper ID

159

Paper Title

Generative AI Models: A Review of Architectures and Challenges

Track Name

Hardware/Software Solutions for AI & Big Data (SAB)

REVIEW QUESTIONS**1. Overall evaluation:**

border line

2. Reviewer's confidence

Medium

3. Relevance of the contribution for IEEE-iSES 2025

appropriate

4. Contribution to the computing education research field

minor contribution, perhaps with the promise of more to come

5. Likelihood of generating discussion at the conference that will benefit the field

neutral

6. Best Paper Recommendation?

No

7. Paper for poster presentation recommendation?

Yes

8. Paper comments:

- ☐ Content and Scope: A variety of generative AI models, such as Transformers, VAEs, GANs, Boltzmann Machines, and Auto- Regressive Models, are reviewed in the study. Futuredirections in the field as well as opportunities and problems are covered. It provides a good introduction to the topic because it covers a wide range of important architectures.
- ☐ Originality: It is a review and does not include new empirical data, new methods, or original research. It limits its contribution to the topic by summarizing the body of literature and information already in existence.
- ☐ Strengths: The essay offers a concise synopsis of the basic architectures and their features and is clearly organized and easy to read. To classify the applications of these models, it is helpful to group them according to input-output modalities (text-to-text, text-to-image, etc.). It also skilfully draws attention to significant computational, security, and ethical issues.
- ☐ Areas for Improvement: There is not enough critical analysis or depth in the review.